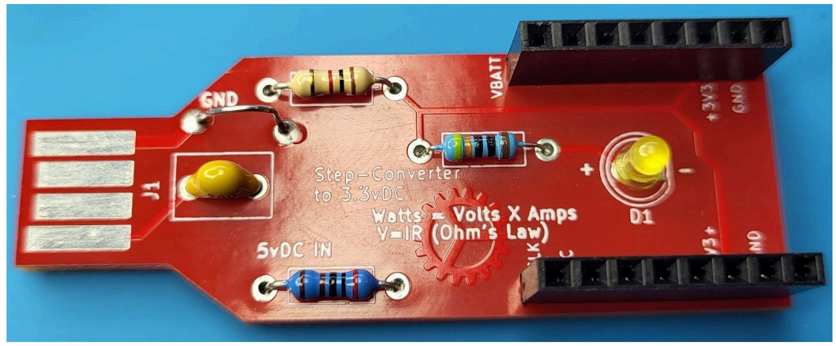
USB Solder Starter Kit

By Josh Kesecker

Materials:



RED & Silver PCB
1 x Ceramic Capacitor
1 x 1K Resistor (Tan)
1 x 2K Resistor (Dark Blue)
1 x 430 Resistor (Light Blue)
1 x Colored LED (Yellow/other)

Help With Soldering?

Checkout the first 5 minutes of this video: https://www.youtube.com/watch?v=3jAw41LRBxU

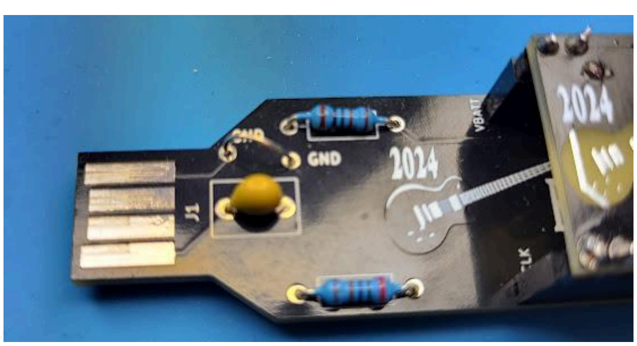
You don't need all their name-brand stuff to get the job done. Soldering irons are seldom used these days, so you can likely borrow one from somebody.

First Assembly: USB Stick

1. Orient the PCB with the component/USB side (front) facing up

2. Take all the resistors, and bend their legs almost 90-degrees so it forms a 'U' shape with the colored component in the middle
3. Insert these long legs through the labeled locations on the top of the motherboard, pushing the components all the way down to the PCB board (direction doesn't matter, but location *does*)
4. Bend the legs on the other side, so the part can't fall out
5. Insert the capacitor in the C1 location (direction doesn't matter)

6. Insert the LED (flat side towards the - symbol), but not all the way
    a. By keeping the LED higher (up to ¼" total height) it'll be closer to minibadge accessories, making them glow better
7. Flip the assembly over, and solder all component legs to the board
8. Reserving 1 (thicker?) leg, trim the rest. (These can fly, so hold them as you cut.)
9. With the spare leg piece, bend in half to form another 'U' or staple-like shape
10. Insert beside the capacitor, in the "GND" holes, with no excess sticking out the bottom
11. You want to solder the legs of this "Ground" loop, so it stands tall on top, for easy probe access

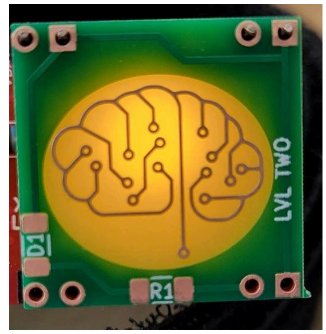


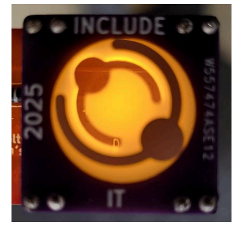
Your USB Stick Motherboard is complete.

TESTING: Do NOT plug this into a computer's USB port at first. Use a Simple/dumb wall charger or other output you care little about.

Plug the PCB directly into a USB-A port. If assembled correctly, the LED should illuminate! (if not, see troubleshooting section at end)

Second Assembly: Mini-Badge Accessories





Materials:

1 x Existing Solder Starter USB board
2 x 8-pin headers
4 x 2-pin pins
1 x Minibadge PCB
(Resistor & LED components vary by badge)

1. Place the 8-Pin headers into the front of your existing USB stick
2. Insert the long ends of the 2-PIN pairs into the corners of those headers
3. Place the minibadge over those 8-pins to align the corners and hold the headers straight up from the board
4. Solder the tops of the 2-pin pairs, thereby affixing them to the minibadge PCB
5. Holding the minibadge components against the board (or using tape/rubberband/etc) flip the entire assembly over to solder the back of the headers
6. Plug the Minibadge back into the USB stick & it should still work, illuminating the back of the minibadge
7. Proceed to solder minibadge components as needed

Troubleshooting:

Is your ground connected where it should be?

The USB stick has labels on the board, where you can find 3vDC or VBATT (5vdc). If you are not getting voltage, check your source or resistors.

Once you verify 3vDC is making it to your header, does it pass to your Minibadge?

Is your LED backwards?